Fiscal Year:	FY 2026	**Task Last Updated:**	FY 11/08/2025
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Project Title:	PeleLM CFD of Ion Driven Winds from Diffusion Flames		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92697	**Congressional District:**	45
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	01/20/2022	**End Date:**	01/19/2027
No. of Post Docs:	1	**No. of PhD Degrees:**	
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No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 01/19/2027 per F. Hernandez/NASA (Ed., 4/8/26). NOTE: End date changed to 01/19/2026 per D. Stocker/NASA GRC (Ed., 2/7/25). NOTE: End date changed to 01/19/2025 per NSSC information (Ed., 3/10/24).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dunn-Rankin, Derek Ph.D. (University of California, Irvine)		
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Task Description:	The ultimate objective of this project is to comprehensively simulate the behavior of a small diffusion flame under the influence of an externally applied electric field in zero-gravity and to use PSI data to confirm the accuracy of that modeling. A detailed model will be used that includes the chemistry of charged ions and chemiluminescent flame intermediates to capture any feedback between ion-driven convection and combustion behavior, and to allow quantitative comparisons with experimental measurements. To date, the capability of accurately simulating electric field flames has eluded researchers because the system exhibits dramatic ranges of coupled temporal and spatial scales. Moreover, in Earth's gravity, the hot combustion products are subject to buoyancy effects that are difficult to isolate from those generated by the electric field. We propose implementations to an existing powerful simulation framework (PeleLM) for this problem, and to use the PSI data set from E-FIELD Flames for validation of the model and investigation of the complex coupled system. PeleLM is a state-of-the-art reacting flow code tailored with a unique integration scheme that efficiently couples high-speed processes with the slower evolution of the flow structures using the adaptive mesh refinement (AMR) strategy. This project is to improve the ion chemistry (particularly employing reduced chemical mechanisms available in the published literature) to generalize the electric field configurations in PeleLM, and to then simulate zero-g coflow flames in the E-FIELD Flames dataset. Comparisons include V-I (voltage-ion current) curves and flame shapes (deduced from CH* images), varying with fuel types, compositions and flow rates. The data includes step function changes in voltage that can be used to evaluate the time response of the flame. It is important to also consider the impact of image and ion current collection timescales, which will be critical for model validation. The work takes advantage of the unique E-FIELD Flames data, measured in microgravity, to validate the proposed PeleLM model augmentations, to permit a detailed investigation of the ion dynamics under the influence of electric fields, and to evaluate the subsequent interactions with non-charged species, as well as to assess the potential of electric field forcing to affect thermal transport and sooting in diffusion flames. Once the simulation is validated against the zero-g data, it will be possible to further extend the simulations to flames in Earth's gravity, where potential applications for improved heat release and emission reduction can be explored. This utilization of zero gravity data from the E-FIELD Flames experiment for beneficial purposes on Earth and for a better understanding of combustion control is compatible with the NASA objectives. The E-FIELD Flames experiment was conceived and directed to create a data set that would permit the detailed assessment of the effects of ion-driven winds and charged species chemistry on non-premixed flames. A microgravity environment is required because the forces of ion driven winds and of gravity driven buoyancy winds are too close in magnitude to effectively separate their effects with any earth-gravity experiments where natural convection is a significant contributor to the combustion process. This rationale has been proven out by the E-FIELD flame experimental data now available as part of the PSI website. The Electric Field Effects on Laminar Diffusion Flames (E-FIELD Flames) experiment was completed in 2018, and with nearly 150 hours of operations over six months and the ignition of 250 electric field flames on the ISS, it was a very successful measurement campaign. The experiment aspect of this project is to extract and assemble a data portfolio that is ideally suited for validating numerical simulations. The simulation aspect of the project employs the PeleLM framework at the National Renewable Energy Laboratory (NREL), which was previously developed in the Lawrence Berkeley National Laboratory (LBNL). This open-source code is for evolving chemically reacting flows at low Mach number with block-structured adaptive mesh refinement (AMR) using the AMReX library. It is the extension of this code that we exploit for comparison with E-FIELD Flames data, with the ultimate goal of understanding and then controlling hydrocarbon flames under the influence of electric fields. PeleLMeX is the non-subcycling version of PeleLM, and this project will use the non-subcycling strategy at the same time understanding other aspects using the PeleLM. Control of combustion using electric fields has for decades been proposed for extending flammability limits, reducing emissions, and preventing instability and blowoff, as well as modifying soot production. Implementation of the concept has been difficult, however, because of the complex coupling between buoyant driven flows, ion production in the flame, ion acceleration by the electric field, and the resulting forces on the neutral gas. With an efficient and accurate simulation capability, it will be possible to explore temporal and geometric variations of the electric field to manipulate flames in a wide range of beneficial ways. With the implementation of a super reduced GRI-Mech 3.0-based mechanism and without excited species and with a few ions, the simulation results did not capture a clear difference between microgravity and Earth gravity conditions. This fiscal year is to continue the previous efforts in implementing the San Deigo-based reduced mechanism, as well as testing the new PeleLMeX code specifically for electric field flames research. The specific aims of this project task include: (1) Understanding the strategy by working with the detailed electric field flames in the PeleLMeX code and its function, including initiating and implementing the code to rapidly perform vast numbers of calculations by graphics processing unit (GPU) (2) Then implementing PeleLMeX CFD with a (newer) reduced chemical mechanism that includes positive ions, negative ions and excited species with 1g condition (3) Verify the fidelity of the PeleLMeX simulation of flames under the influence of electric fields by direct comparison using the previously validated reduced mechanism without electric field in comparison to the schlieren imaging under 1g conditions (4) Demonstrate validated PeleLM simulations of microgravity flames in steady conditions and during unsteady transitions in response to changes in the external electric field
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Once the simulation is validated against the zero-g data, it will be possible to further extend the simulations to flames in Earth gravity, where potential applications for improved heat release and emission reduction can be explored. This utilization of zero gravity data from the E-FIELD Flames experiment for beneficial purposes on Earth and for a better understanding combustion control is compatible with the NASA objectives.

Task Progress:

As in the above objectives listing, this year has major aspects of working in the PeleLMeX computing by establishing the appropriate electric field flames code, with the more complicated San Diego-based mechanism conducting with the International Space Station (ISS) conditions. The current year's effort includes continuing to modify the mechanism (with charged and excited species) and addressing/fixing the details in the PeleLMeX CFD code for simulating flames under the influence of an electric field. This year, progress has been made on all the items with emphasis on 1-3 for the new mechanism. Specifically, the progress includes:

Computational Modeling Implementation Employed with a Set of Species

To predict ion concentrations in flames and the effects of ion driven winds in coflow and jet flames, a complete set of charged species and excited species to include in the computation is essential and the most challenging. This task is to find the best scientific method to include more charged species and excited species implementation into the CFD simulation with appropriate reduced chemistry to ensure robust simulation capability.

The previous PeleLM simulation computation capability validation using pressurized conditions also continues. Further details of the project progress through the fourth year are described in the following.

• PeleLMeX Computation

The progress in the PeleLM computational aspects of the project are described in a series of conference papers provided in the publication section of this report and in recent American Society for Gravitational and Space Research (ASGSR) presentations (Grabon et al., 2024, Deu Morel et al., 2023). Much of the work has so far concentrated on the cases without an electric field in order to assess the complications of the code implementation, its conversion to cylindrically symmetric geometry, to identify the choice of reduced models, and to ensure that the boundary conditions are appropriate to simulate an extruded fuel tube.

• Implementation with more species in PeleLMeX – Preliminary Outcome

The PeleLMeX computation of this research is implemented on the UCI High Performance Community Computing Cluster (HPC3) running on Linux 8.6. Its aim is to boost the effectiveness of simulations by utilizing a considerable number of cores. To handle user admission and resource allocation, the cluster employs entities like login nodes and the Slurm scheduler.

As this is the second set of the reduced mechanism for the new PeleLM - PeleLMeX code, the goal of this part of the project is to use the reduced mechanism tested for microgravity conditions. This work initiated with reproducing the behavior of the coflow burner used in the ISS. The primary distinction between the numerical and experimental setups lies in the extruded fuel pipe. As the PeleLMeX code has not integrated the extruded geometry, it is unable to replicate the impact of the fuel pipe on the inflow velocities or temperature. The effect of the extruded tube is simulated using a customized inlet velocity profile. The primary assumption is that the Low Mach number regime is in effect. It is acknowledged that the velocities will remain within this regime. This is a widely accepted hypothesis in the study of combustion and was validated at the conclusion of the simulations.

The simulation was initiated with 19 (and 15) cm/s fuel flow velocity with 3.1 cm/s coflow air with 300K initial temperature with CFL 0.1. This case without the electric field can help explain the operation of PeleLMeX. Moreover, the cases without electric field are distinctive for its shorter compilation time due to the reduced number of equations/species to be solved resulting from the deactivation of the electric field. To bridge the gap between simulation outcomes and empirical data, this part of the work is to evaluate potential mechanism of 45 species and 216 reactions using San Diego reduced mechanism, by López-Cámara et al. This task is to implement the recently published reduced San Diego mechanism in microgravity and with an electric field. These mechanisms are chosen for their comprehensive coverage of ionic and exited species within the flame compared to the mechanism currently used by Di Renzo et al. which is not able to capture the ion current behavior in microgravity condition.

The main difference lies in the reduced San Diego mechanism chemical model, which includes 11 charged species, including H3O+, HCO+, C2H3O+, CH5O+, O2−, OH−, e−, CO3−, CHO2−, O−, and CHO3− (in comparison to the previous used H3O+, HCO+, O2− and e−) and included excited species CH and OH. This mechanism includes many species, and as the electric field is introduced the calculation became extensive and the computation time increased substantially.

With the numerous attempts in the PeleLMeX, the heat released from the calculation is massive in the prior year; this year we have initiated the computation from a GPU desktop and also the HPC supercomputing on campus. The GPU desktop is useful for locally operating the PeleLMeX simulation, and then further allowing the migration to a more complicated environment in the HPC system. This fiscal year, with the PeleLMeX code initiator (Dr. Day's help) in fixing the San Diego-based PeleLMeX, the code was successfully ignited with a flame at 70% methane concentration at 15 cm/s under 1g conditions. The flame reaches the steady state after several days of running via UCI's HPC and with the GPU code embedded into the PeleLMeX code.

All testpoints are ignited with a blob with a radius of 1 mm and a temperature of 1,900 K (based on previous work), which ensures an ignition in all cases. The beginning of the ignition being the most unstable part, the simulation was run with a low CFL number of 0.05 in order to have the most stable run, which also meant computation time was quite high. However, even with this setting, stability issues arose: simulation failed around 0.2 ms as the solver was unable to compute the ignition due to initial conditions. To fix this issue, the simulation was set to cap the time-step at 10 - 7 at the beginning of the ignition. In this configuration, the simulation can run with CFL=0.08, while remaining stable. After a run of 0.03 s, the time-step limit was switched off and simulation could run with a time-step of 10 - 5 until reaching steady state. During those first 0.03 s, a set of species at the initial state was used to prevent computation error without the essential species. The current project continues with four testpoints selected before and after the acme point, and they are -3.7, -4.4, -5.5, and -7.4kV.

Conclusions

The newer version of PeleLM (PeleLMeX), using the recently published reduced mechanism, is progressing. Specifically, the non-subcycling version of PeleLM has been debugged and implemented with a GPU. The code is running with the flame ignited. These tests demonstrated that the code will take longer on a weekly basis to run for milliseconds. The code also needs to be continued to be exercised under different coflow flame conditions with different levels of fuel dilution, with carbon dioxide and different pressures, to demonstrate the validity of the previous work with

	water addition to the boundary condition implementation. Challenges remain for how this relatively large San Diego-based mechanism will be able to accomplish all of the testpoint tests, including more ions and charged species into the newer version of PeleLMeX; and further, whether it will capture the ion current curve measured from the ISS. Continuing work on this project focuses more heavily on the computations, with a detailed assessment of the ion current prediction and the role the chemical mechanism plays in that prediction. References Deu Morel, R., Day, M. and Chien, Y.-C. (2023) “Investigation of mechanisms for microgravity ion-driven wind using simulation,” Graduate Poster Presentation, American Society for Gravitational and Space Research Annual Meeting 2023, Nov. 14 –18, Washington DC. Grabon, K., Wimer, N., Deak, N., Day, M, and Chien, Y.-C. (2024) “Ion-driven wind of E-FIELD Flames,” Combustion 2 Presentation, American Society for Gravitational and Space Research Annual Meeting 2024, Dec. 3 – 7, San Juan, Puerto Rico.
Bibliography Type:	Description: (Last Updated: 11/10/2025)
Dissertations and Theses	Le Moal, B. "Mechanism with ions and excited species implementation for simulating E-FIELD Flames under microgravity influences." M.S. Thesis for Engineering Internship from ISAE-ENSMA, University of California, Irvine (in progress). , Dec-2025